

Evolution of Safety Concepts in Pressurized Water Reactors

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I. INTRODUCTION

Nuclear power plant safety remains a crucial concern in present society, generating massive attention of both the public and experts in the field. Continuous advancements in reactor design and safety systems emerged over the span of different generations, integrating state of the art technology, and adjusting to the evolving regulatory requirements. This study meticulously examines and compares safety systems across three generations of pressurized water reactors: Generations II, III, and III+ (see Table 1).

TABLE 1: ANALYZED REACTOR TYPES OF DIFFERENT GENERATIONS.

Generation	Reactor Type
II	KWU
III	APR-1400
III+	EPR, AP1000

A. Abbreviations and Acronyms

CFS.....Cavity Flooding System
 CSS.....Containment Spray System
 DBA.....Design Basis Accident
 DCH.....Direct Containment Heating
 ECCS.....Emergency Core Cooling System
 EFWSEmergency Feedwater System
 ERVCSExternal Reactor Vessel Cooling System
 IRWST.....In-Containment Refueling Water Storage Tank
 NPP.....Nuclear Power Plant
 KWUKraftwerk Union AG
 LB.....Large-Break
 LOCALoss-Of-Coolant Accident
 LOFW.....Loss-Of-Feedwater
 MSLBMain Steam Line Break

RCS Reactor Coolant System
 RHRS Residual Heat Removal System
 RPV Reactor Pressure Vessel
 PWR Pressurized Water Reactor
 SCS..... Shutdown Cooling System
 SDVS..... Safety Depressurization and Vent System
 SG..... Steam Generator
 SIS Safety Injection System

II. METHODOLOGY

A literature review has been selected as the research methodology for this paper. In the process, more than 70 sources have been analyzed and compared. Key information regarding the development and evolution of safety systems of PWRs has been meticulously collected and provided the basis for a comparison of key safety designs and systems.

III. FUNCTIONALITY AND GENERAL SAFETY CONCEPTS

B. Functionality of a PWR

A PWR can be separated into two systems: The primary and the secondary system. The main components of the primary system include the reactor pressure vessel (RPV), steam generators (SG), the reactor coolant pumps, the pressurizer and the piping, as well as the safety-related systems which will be described in more detail in the following chapters. The primary system is a closed loop and has the main task of cooling the core and transporting the heat away from the fuel to the steam generators, where the heat is transferred to the secondary system. The primary system is enclosed by a containment structure for all discussed reactor types. The secondary system consists of the main steam system and the condensate/feedwater system, which are both located outside the containment building at the turbine building [1].

C. General Safety Concepts

The most important part of nuclear safety relies on the supposition that the on-site staff of a NPP as well as the public health in general is always protected from the harmful effects of ionizing radiation [2]. The safety philosophy of “Defense in Depth” is an integral part of the design-process, operation and maintenance of nuclear facilities. Furthermore, the objective of this approach includes protecting the environment while securing the functionality of the plant. “Defense in Depth” is achieved by the following design and safety principles [2], [3]:

Multiple Barriers: Multiple layers of safety barriers prevent the release of radioactive material. The most important barriers can be seen in Figure 1 and consist of the (1) structure of the fuel, (2) fuel cladding, (3) RPV, (4) biological shield, (5) containment and (6) reactor building. [3]

Redundancy and Diversity: For important safety-related mechanisms, there are multiple systems installed which serve the same purpose (redundancy). These systems do not only work independently, but also achieve their desired task by different ways to minimize the danger of common-mode failures (diversity). [2]

Physical Separation: Besides the existence of multiple backup mechanisms, those systems are also separated physically. This ensures that e.g., in the case of a fire or flooding of an area, not all systems that serve a specific purpose are destroyed. [3]

Fail-Safe-Principle: The design of relevant safety components should ensure that in case of an emergency, the reactor transitions by itself into a safe state. E.g., in case of a power outage, the control rods are inserted into the core by gravitational force and thus the nuclear reaction ceases. [3]

Automation: In case of a safety relevant incident or accident, the reactor protection system is designed to counteract and initiate safety measures automatically without the need of an active intervention by working personnel for at least 30 minutes. This mitigates the risk of human errors under significant pressure. [3]

Self-Sufficiency: The functionality of vital safety systems (e.g., the residual heat removal of the core after shutdown) has to be ensured for an extended period of time, even in case of a loss of electrical power. This is achieved by backup power supply systems like a diesel generator. [3]

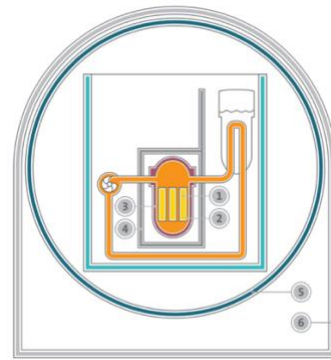


Figure 1: Multiple safety barriers to prevent the release of radioactive material. [4]

IV. SAFETY CONCEPTS OF GENERATION II REACTORS

In pursuit of a safety concept, which suffices the western, and particularly German nuclear safety standards, the Kraftwerk Union AG (KWU), a subsidiary company of Siemens and at the time AEG, designed the KWU-reactor with safety measures in mind which are still present today in modern pressurized water reactors. The three reactors which were analyzed in more detail were the NPP Biblis A and B, as well as NPP Grohnde, which are now all in decommissioning or shutdown. The safety concept of this reactor type can be divided into three isolated systems: **Passive-, active- and reactor protection system**. The main design basis accidents (DBA) include large-break loss-of-coolant accidents (LBLOCAs), although the possibility of a main steam line break (MSLB) is also considered. [5] A rupture of the RPV was not considered as a DBA, and although the RPV itself is not redundant, the different approaches regarding the quality of production and monitoring can be considered as diverse and redundant. [6] The severe accident scenario of a core meltdown was not considered in the design process. Therefore, no safety systems were integrated which would mitigate a core melt accident and potential rupture of the RPV. Through gained knowledge and experience, the lack of dedicated safety systems for accident scenarios not formerly considered prompted improvements and major innovations of safety designs and systems, resulting in Generation III and III+ reactor designs. These technological leaps gained immense importance after the accidents of Chernobyl in 1986 and Fukushima in 2011, when the nuclear industry came under scrutiny and had to prove that future reactors will withstand extreme scenarios and are a safe source of energy generation.

D. Passive Safety Systems

The containment structure poses the most important passive safety system of the KWU reactor. It consists of a highly gas-tight inner containment, a ventilated annulus with a filter system, which is connected to a chimney and serves as a suction in case of a leakage, and an outer containment.

The outermost layer is a cylindrical, gas-tight structure made of reinforced concrete. The containment houses the reactor coolant system (RCS), the steam generators (SG), the spent fuel pool and the crane. In case of a loss-of-coolant accident (LOCA), separated parts could damage important components of the cooling circuit and are therefore enclosed by reinforced concrete. [7] The inner containment is designed to withstand a pressure increase of up to 5.7 bar, induced by a LBLOCA. It is assumed that the total content of the RCS leaks into the structure and that the increase of pressure is caused primarily by the decay heat of the core. [8] The safety philosophy of western countries regarding nuclear power, including Western Germany, was at that time different to the approach of the Soviet Union. With the beginning of the commercialization of nuclear power, the implementation of a containment was required for a licensing permit in Western Germany. The Generation II RBMK reactor, originating from the Soviet Union, does not feature a containment structure, nor was it formally required in the deployed countries. The Chernobyl disaster in 1986, inter alia, highlighted the importance of the lacking containment structure, as large quantities of radioactive material were released directly into the environment and spread over large parts of the former Soviet Union and Europe.

E. Active Safety Systems

In case of transient or accident conditions, the reactor protection system initiates various active safety systems to minimize the resulting consequences and radiological danger. [8] With the second generation of NPPs, a variety of active systems were designed and implemented to prevent and mitigate DBAs, as well as other transient and accident conditions. While these systems are effective when they are available, maintaining availability and functionality at all times proves challenging over the operational lifetime due to their active nature, particularly in combined stress situations. The residual risk, in German termed “Restrisiko”, of a severe accident with a large release of activity has preoccupied engineers in the western world since the early days of the commercial use of nuclear power in the 1970s. In order to quantify the involved risks with a probabilistic risk assessment and analysis approach, the German risk studies stage A and B were composed. Based on the results of these studies, which were published in 1979 and 1990 respectively, various improvements of safety related systems were undertaken [7, p. 674]. The discussion about the risk of nuclear power has not been limited to the scientist community but was also widely discussed in the German public. After the Fukushima Daiichi accident in 2011, the discourse once more gained immense momentum and partially contributed to the shutdown and phasing out of German nuclear power. The KWU-PWR features several at the time cutting-edge safety systems, which can still be found in today’s newest reactors:

Emergency Core Cooling System: In case of a loss-of-coolant accident (LOCA), the reactor protection system activates the emergency core cooling system (ECCS), which consists of different injection systems. [8] These separate systems comprise high to low pressure injection trains to counteract different pressure variations in the RCS following primary breaks. The pumps of these systems direct borated water from a dedicated refueling water storage tank into the primary circuit. For LBLOCAs and therefore a large pressure drop in the RCS, the cold leg accumulators are designed to supply large amounts of borated water from their tanks. This system works passively, without the need of electrical power, by a nitrogen gas bubble that is enclosed on top of the tank. Additionally, the residual heat removal system (RHRS), which acts as a low-pressure system, is actuated. This system also pumps water from the refueling water storage tank into the RCS. At some point, the two tanks are drained and the RHRS pumps the water from the containment sump through the RHRS heat exchanger, which extracts the remaining heat and directs it back to the core. Additionally, the KWU-PWR features an emergency feedwater system (EFWS), which supplies water from the condensate storage tank when the main feedwater system does not supply the necessary feedwater to the SG. This ensures the necessary heat extraction from the RCS. [1]

Hydrogen Monitoring and Control: Following a LOCA, the supplied coolant from the emergency system undergoes a radiolysis process due to the ionizing radiation of the core and spent fuel: $H_2O \rightarrow H + OH$. If the temperature of the fuel cladding, typically made of a zirconium alloy, reaches 1200 °C, the greatest contribution of hydrogen production results from the zirconium-water-reaction: $Zr + H_2O \rightarrow ZrO_2 + 2 H_2$. [8] The formed hydrogen accumulates in the containment, and the danger of a detonation or deflagration arises if the H_2 -concentration increases. This process also depends on the oxygen and steam concentrations within the containment atmosphere. To prevent such an event, the KWU-PWR features a hydrogen control system, consisting of a ventilation system, that circulates the air inside the containment and therefore prevents high local hydrogen concentrations, and hydrogen recombiners. [8]

V. SAFETY CONCEPTS OF GENERATION III REACTORS

Generation III NPPs are a further development of Gen II reactors. The implemented evolutionary design improvements in the areas of safety and economics are based on the gained operational experience from the previous generation. For this section, the Advanced Power Reactor 1400 (APR-1400) has been selected, which was realized at two sites in the Republic of Korea, as well as at the Barakah NPP in the United Arab Emirates. [9] The systems designed to deal with DBAs have been further developed with an emphasis on passive working principles and simplicity, in order to increase availability and

reliability, as well as to reduce the cost of the deployed safety systems. This reactor features novel and innovative safety systems, designed for severe accident scenarios (beyond-DBAs), which can be divided into two categories:

F. Severe accident prevention

This includes increased design margins, e.g., a higher thermal margin, as well as the integration of systems like the safety injection system (SIS), the in-containment refueling water storage tank (IRWST), the containment spray system (CSS) and the safety depressurization and vent system (SDVS). [10], [11]

The SDVS provides a countermeasure for high-pressure core melt scenarios by depressurizing the RCS [10]. This scenario was not considered for the Gen II KWU plants but gained more attention after the Three Mile Island Unit 2 (TMI-2) accident. In case of a high-pressure core melt, the RPV could fail and lead to direct containment heating (DCH). The SIS is designed to inject water directly into the core to remove decay heat following a LOCA and comprises four trains. For each train, there is one safety injection pump (SIP) and one safety injection tank with an integrated passive fluidic device.

The fluidic device works based on vortex flow resistance and regulates the flow rate of the borated water passively, in order to optimize the water injection rate following a LOCA. This eliminates the need for a low-pressure SIP. The different operation modes used in older NPPs can be replaced by one operation mode (safety injection mode). [11] The residual decay heat removal after a reactor shutdown is provided by a different, dedicated system called the shutdown cooling system (SCS), along with the EFWS. It can also be used to cool down the RCS in case of a SBLOCA. [12] The simplified design of the SIS and the separation of the initial emergency cooling by the SIS and the residual heat removal by the independent SCS increases the reliability and availability of the related safety systems compared to the arrangement used in Gen II KWU plants, where the RHRS is part of the low-pressure injection system of the ECCS.

The CSS is designed to reduce the temperature and pressure inside the containment after a LOCA or MSLB by spraying borated water from the top of the containment. Two independent trains deliver the water taken from the IRWST through dedicated containment spray pumps. These pumps can be used as backup pumps for the SCS to increase redundancy and reliability. [10]

G. Severe accident mitigation

These systems are designed to limit the possible consequences and the damage caused by a major accident (beyond-DBA), in particular to prevent an early containment failure. [10] They, inter alia, include the robust containment, the hydrogen mitigation system, the cavity flooding system (CFS)

and the external reactor vessel cooling system (ERVCS). Compared to Generation II NPPs, the deployment of the CFS and the ERVCS increases the probability that the containment does not fail in case of a severe accident. Securing the integrity of the RPV and consequently the containment presents an integral aspect in safeguarding the principle of barriers and thus limiting the possibility of a radioactive release beyond these boundaries.

The ERVCS is an active system ensuring the in-vessel retention of the corium in case of a severe accident involving a core meltdown by filling the reactor cavity up to the level of a hot leg and supplying enough water to keep the cavity flooded, as the decay heat evaporates the initial coolant. Probabilistic safety assessments of PWRs strongly indicate that the retention of the corium inside the RPV and therefore ensuring the integrity of the RPV presents a critical role in safeguarding the intactness of the containment. [10]

In the severe accident scenario of a core meltdown and consequently the formation of corium, the reactor cavity is specifically designed to contain the corium in the core debris chamber and to prevent DCH through the convoluted cavity design and the CFS. The CFS works passively by harnessing gravitation and comprises two trains which connect the IRWST to the reactor cavity via the hold-up volume tank (HVT) (see Figure 2). [13] the resulting steam from the cooling of the core debris is then condensed by the CSS. The CFS prevents interaction between the molten corium and concrete. It furthermore averts the formation of combustible gases and reduces fission product releases. The CFS is intended to be used as a backup for the ERVCS in case the in-vessel retention of the corium has not been successful and the RPV has failed. [10], [13]

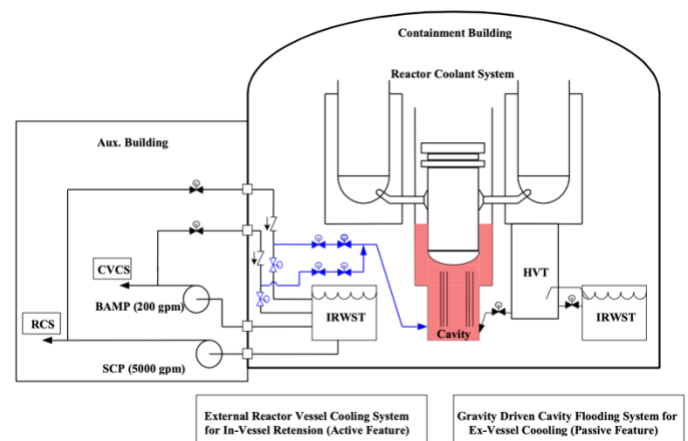


Figure 2: External reactor vessel cooling and cavity flooding system of the APR-1400. [12, p. 44]

VI. SAFETY CONCEPTS OF GENERATION III+ REACTORS

The development of Gen III+ plants started in the 1990s. Their design is very similar to Gen III reactors, with further emphasis on safety systems that rely on passive operating principles like natural convection cooling and gravity driven injection of water rather than solely on active safety elements. [9] In order to deal with severe accidents (beyond-DBAs), these reactors feature unique safety systems and designs which were developed based on the experience accumulated through the decade-long operation of light water reactors combined with extensive safety analyses. The aim is also to further minimize any grave radiological consequences of severe accidents. Two systems of this generation will be discussed in this chapter more thoroughly. Different Generation III+ reactors generally vary in their technical design, but the safety systems are based on similar passive working principles to ensure the highest achievable availability and reliability.

H. European Pressurized Reactor

The European Pressurized Reactor (EPR) is based on the German Konvoi and French N4 reactor designs and has a thermal power output of around 4500 MWth. The reactor was built at the Taishan NPP (China), Olkiluoto NPP (Finland) and Flamanville NPP (France) and is under construction in the UK. The EPR features novel approaches to deal with the severe accident scenario of a partial or total meltdown of the core, as the formed corium could lead to a high-pressure core melt resulting in DCH, or inter alia, to a low-pressure core melt penetrating the containment basemat. [14] Because of the high power rating of the EPR, a different design approach had to be selected than for the APR-1400 as the in-vessel retention of the corium by ex-vessel cooling is expected to be very unlikely. The EPR features a sophisticated core catcher design, which retains and cools the corium in the spreading compartment passively. This is achieved with water flowing from the IRWST by gravitational force (see Figure 3). The cooling water is subsequently evaporated by the decay heat and afterwards condensed by sprays. The containment heat removal system (CHRS) works actively and ensures that the pressure inside the containment is limited while maintaining the integrity of the structure in the long-term. The EPR also features a depressurization system, designed to prevent severe accident scenarios of a high-pressure melt-through of the RPV or a steam generator tube rupture, which could lead to DCH or a containment bypass. This is achieved by depressurizing the RCS with dedicated valves. [14]

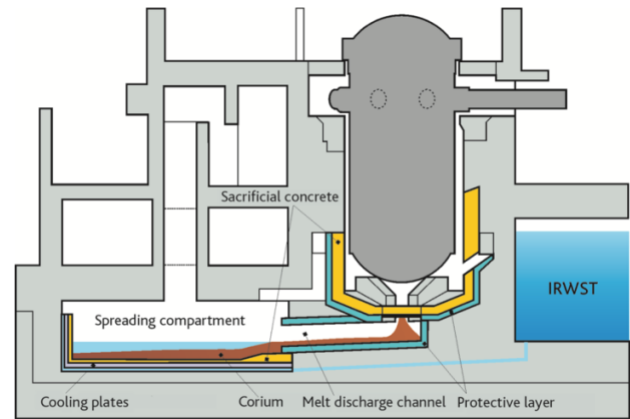


Figure 3: Cutaway view of the EPR core catcher. [14, p. 85]

The EPR utilizes a double-wall containment with an annulus ventilation system. It is similar in its layout to the containment used in Gen II KWU plants. The containment design of the EPR has been upgraded to ensure that no significant uncollected leakage past both containment layers occurs for both LOCAs and core melt accidents. [14]

I. Advanced Passive Reactor 1000

The AP1000 incorporates extensive passive design elements and safety systems, harnessing natural forces like gravity and natural convection for the most important safety systems and eliminating the need for major active components and action by plant personnel. The containment design incorporates a passive containment cooling system, which enables the removal of the decay heat from the RCS by natural convection and gravity following DBAs as well as core melt accidents. The passive core cooling system comprises the passive RHRS and the safety injection and depressurization system to provide emergency core cooling and residual heat removal in various accident scenarios. The injection of borated water is driven by gravity. The passive RHRS heat exchanger utilizes the IRWST as a heat sink to fulfill the necessary cooling in case the secondary side heat removal is not available. For the severe accident scenario of a core melt, the strategy of in-vessel retention through external reactor vessel cooling has been selected. The utilized system is similar to the ERVCS used in the APR-1400. The AP1000 does not feature an ex-vessel retention and cooling system as a backup. [15]

VII. CONCLUSION

With the Generation II KWU plants, the “Defense in Depth” approach to nuclear safety has been incorporated successfully: Both passive and active safety systems ensure that in case of the regarded DBAs, mainly LOCAs and loss-of-feedwater (LOFW) accidents, major radiological consequences can be prevented or mitigated. The use of a double-layer containment with a ventilated annulus combined with the containment isolation and

hydrogen control offers advantages in containing the release of radioactive material in case of a severe accident compared to Gen II NPPs, which do not feature these safety structures and systems. The active safety systems, including the ECCS, the EFWS as well as the containment isolation, present a balanced net of safety to counteract DBAs, especially LBLOCAs and LOFW accidents. The emergency power supply of these systems is designed by considering both redundancy and physical separation. A weakness of the active safety systems is their relative high complexity and active nature, leading to vulnerability in combined stress situations. The lack of safety systems preventing and dealing with beyond-DBAs like a core meltdown is also a clear drawback compared to subsequent Generations.

With the Generation III APR-1400, design improvements were made in the areas of safety and economics. The latter is achieved with a reduced complexity of major systems and with an increased design margin. These changes increase the reliability and availability of the systems compared to the Generation II KWU plants. The IRWST is located inside the containment for better protection against external forces and provides borated water to the safety systems. Additionally, a CSS is installed inside the containment for the purpose of reducing the containment pressure and temperature after an accident. Compared to the Generation II KWU plants, major advancements were made in the mitigation of core melt accidents: The strategy of in-vessel retention of the molten corium has been selected. The SDVS is able to reduce the pressure of the RCS, preventing a high-pressure core melt, while the ERVCS aims to cool and retain the molten corium inside the RPV. For the possibility of a RPV-failure, additional provisions were made with the robust cavity design and the CFS.

The Gen III+ EPR features several design approaches to prevent early and late containment failures and thus any grave radiological release following a severe accident. The EPR is equipped with a depressurization system, similar to the APR-1400's. The EPR's safety systems for severe accident scenarios are specifically designed to accommodate the reactor's high thermal rating, with particular emphasis on the sophisticated core catcher design featuring passive cooling.

The AP1000 follows a more drastic and innovative design route by utilizing natural forces like gravity and natural convection for major safety systems preventing and mitigating severe accidents, including the containment cooling and core cooling system. The deployment of the discussed passive safety systems of the AP1000 provides advantages regarding the self-sufficiency of these arrangements. Furthermore, the advantages result in an enhancement regarding the prevention and mitigation of both DBAs and beyond-DBAs.

There is a clear need for a higher share of low-carbon energy sources in the global energy mix. With the rising construction costs and construction time of Generation III and especially Generation III+ reactors, nuclear power has to prove its economic competitiveness on the open market. The construction projects of the EPR and AP1000 in Europe and the USA have deepened these concerns, as they have been over-budget and over-due. At the same time, the successful commissioning of APR-1400s in the UAE and Korea, as well as EPRs and AP1000s in China has demonstrated that Gen III/III+ nuclear power projects can be practically realizable on a tighter budget and timeframe, and thus be economically feasible. This indicates that nuclear power is a viable source of low-carbon energy for the future.

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